Causal Inference

Lab9: Panel Methods – Fixed Effects and Difference-in-Differences

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Topics

1. Fixed Effects and DiD
2. Staggered DiD and the Problem with TWFE
3. Event Study Designs
4. Modern DiD Estimators
5. Factorial DiD
6. DiD with Continuous Treatment

TWFE and DiD

- The classic 2×2 DiD as a regression:

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}$$

- α_i (unit FE) absorb level differences; λ_t (time FE) absorb common shocks
- What remains is $\hat{\tau} = (\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}}) - (\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}})$
- With covariates: $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + X_{it}'\beta + \varepsilon_{it}$
  - Only *time-varying* covariates matter (time-invariant absorbed by α_i)
  - Avoid bad controls; cluster SEs at the unit level

Staggered DiD and the TWFE Problem (Goodman-Bacon, 2021)

- With staggered treatment, TWFE $\hat{\tau}$ is a weighted average of all 2×2 DiD estimates: $\hat{\tau}^{TWFE} = \sum_k w_k \hat{\tau}_k$
- Comparisons include:
  1. Early/late-treated vs. never-treated (good)
  2. **Early-treated vs. late-treated** and vice versa (problematic!)
- Already-treated units serve as "controls" — but their outcomes reflect treatment!

Negative Weights in TWFE

- Weights w_k depend on group sizes and variance of D_{it}, **not** on what we care about — some can be **negative**
- Consequence: $\hat{\tau}^{TWFE}$ can be negative even if *every* unit has a positive effect
- Intuition: if effects grow over time, early-treated "controls" have rising outcomes $\rightarrow$ comparing late-treated to them biases estimates downward
- Takeaway: TWFE is fine **only if** treatment effects are constant across groups and over time

Event Study Designs

- An event study lets us examine *dynamic* treatment effects
- Define *relative time*: $k = t - G_i$, where G_i is the treatment adoption date for unit i
- The event study regression:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k \cdot \mathbf{1}[t - G_i = k] + \varepsilon_{it}$$

- τ_k: treatment effect at k periods relative to treatment
- $k = -1$ is the reference period (normalized to zero)

Interpreting the Event Study Plot

- Pre-treatment coefficients ($k < 0$): test for parallel trends
  - Should be close to zero and statistically insignificant
  - If not $\rightarrow$ parallel trends assumption is suspect
- Post-treatment coefficients ($k \geq 0$): dynamic treatment effects
  - Immediate vs. gradual effects
  - Fade-out or persistence
- Caveat: pre-trends test has low power (absence of evidence $\neq$ evidence of absence)

Problems with TWFE Event Studies

- The same negative-weighting problem from TWFE applies here!
- Sun and Abraham (2021): TWFE event study coefficients are contaminated
  - Each $\hat{\tau}_k$ is a weighted average across cohorts
  - Weights can be negative $\rightarrow$ pre-trends can look "significant" even when parallel trends holds
- This means we can't trust either:
  1. The pre-trend test (false rejections)
  2. The dynamic effect estimates (biased)

Modern Solutions

- Several new estimators fix the TWFE problem. Key idea: **avoid using already-treated units as controls**
- Callaway and Sant'Anna (2021)
- Sun and Abraham (2021)
- de Chaisemartin and D'Haultfœuille (2020)
- Borusyak, Jaravel, and Spiess (2024)

- All share the same core principle; differ in implementation details

Which Estimator to Use?

- If treatment timing varies across units $\rightarrow$ use a modern estimator
- Callaway & Sant'Anna: most flexible, allows covariates, different control groups
- Sun & Abraham: convenient within `fixest` framework
- de Chaisemartin & D'Haultfœuille: allows treatment to switch on and off
- Practical advice:
  1. Run TWFE as a baseline
  2. Run Bacon decomposition to diagnose problems
  3. Use a robust estimator as your main specification
  4. Report both for transparency

No Clean Control? Factorial DiD

- Xu, Zhao, and Ding (forthcoming, JASA)
- A framework for settings where an **event affects all units**
- Setup: $G_i \in \{0, 1\}$ (baseline factor), $Z_i \in \{0, 1\}$ (universal exposure post-event)
- Key question: how does G modify the effect of Z?
  - E.g., does social capital (G) affect response to famine (Z)?
- Challenge: all units exposed post-event $\rightarrow$ no "pure control"
- Two distinct estimands:
  - **Effect modification** (τ_{em}): compares effects *across observed subgroups* $\tau_{\text{em}} = \mathbb{E}[\tau_{i,Z} \mid G_i = 1] - \mathbb{E}[\tau_{i,Z} \mid G_i = 0]$
  - **Causal moderation** (τ_{cm}): causal effect of changing G on the effect of Z $\tau_{\text{cm}} = \mathbb{E}[Y_i(1,1) - Y_i(1,0) - Y_i(0,1) + Y_i(0,0)]$

Factorial DiD: Identification

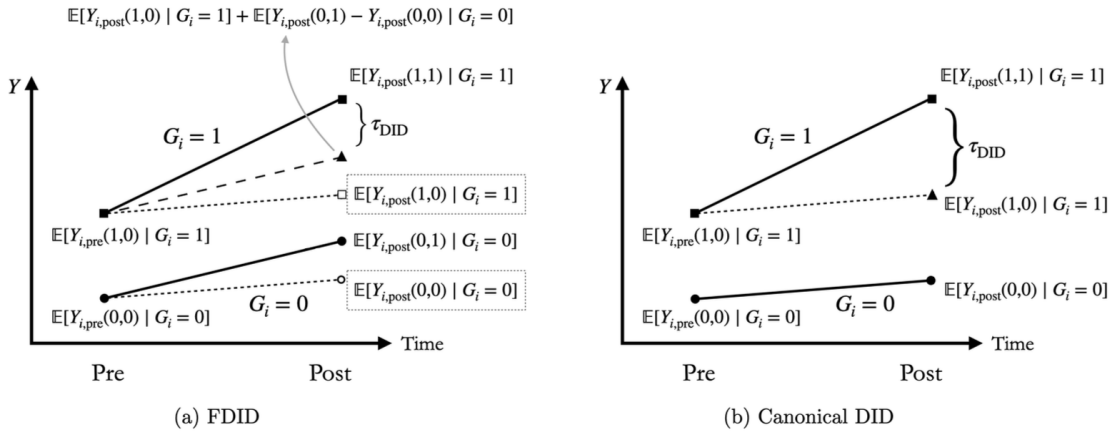

Figure 3: Identification under the no anticipation and parallel trends assumptions.

1. **Universal exposure**: $Z_i = 1$ for all i post-event
2. **No anticipation**: G does not affect outcomes before exposure
3. **Parallel trends**: $\mathbb{E}[Y_{i,\text{post}}(0,0) - Y_{i,\text{pre}}(0,0) \mid G_i = 1] = \mathbb{E}[Y_{i,\text{post}}(0,0) - Y_{i,\text{pre}}(0,0) \mid G_i = 0]$
4. **Exclusion restriction** (canonical DiD): $\mathbb{E}[Y_{i,\text{post}}(0,1) \mid G_i = 0] = \mathbb{E}[Y_{i,\text{post}}(0,0) \mid G_i = 0]$; G has no direct post-event effect on unexposed outcomes
5. **Factorial parallel trends** (for causal moderation): $\mathbb{E}[\Delta Y_i(g,z) \mid G_i = 1] = \mathbb{E}[\Delta Y_i(g,z) \mid G_i = 0]$ $\forall\, g, z$; stronger than standard PT — enables $\tau_{\text{DID}} \to \tau_{\text{cm}}$

- 2, 3, and 5 $\to \tau_{\text{em}} = \tau_{\text{cm}}$
- 1-3, and 5 $\to \tau_{\text{DID}} = \tau_{\text{em}} = \tau_{\text{cm}}$

Factorial DiD: Identification and Takeaways

- Under standard parallel trends: DiD identifies **effect modification** τ_{em}, **not** causal moderation
- To identify τ_{cm}, need **factorial parallel trends** (stronger):

$$\mathbb{E}[\Delta Y_i(g,z) \mid G_i = 1] = \mathbb{E}[\Delta Y_i(g,z) \mid G_i = 0] \quad \forall g, z$$

  Intuition: trends would be the same even under counterfactual G values
- Regression: standard TWFE recovers τ_{did}:
  $Y_{it} = \alpha_i + \lambda_t + \tau \cdot G_i \cdot \text{Post}_t + \varepsilon_{it}$
- Canonical DiD is a special case (when G = treatment + exclusion restriction: $\mathbb{E}\left[Y_{i,\text{post}}(0,1) \mid G_i = 0\right] = \mathbb{E}\left[Y_{i,\text{post}}(0,0) \mid G_i = 0\right]$)
- **Takeaway**: be clear about whether you estimate effect modification or causal moderation — they answer different questions and require different assumptions

DiD with Continuous Treatment: The Problem

- Many settings have continuous/multi-valued treatment (e.g., minimum wage level, exposure intensity)
- TWFE with continuous D_{it}: $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}$
- Problem: $\hat{\tau}$ suffers from **negative-weighting issues** — even worse than binary case
  - TWFE implicitly compares units with different doses; no clear "control group"
  - Some weights are negative even under constant treatment effects (de Chaisemartin & D'Haultfœuille, 2023)

Solution: Callaway, Goodman-Bacon, and Sant'Anna (2024)

- Define causal effects relative to a reference dose ($d = 0$):
  $ATT(d \mid d) = \mathbb{E}[Y_t(d) - Y_t(0) \mid D = d]$
- Summarize via the *average causal response on the treated*:
  $ACRT = \mathbb{E}\left[\frac{\partial}{\partial d} ATT(d \mid d)\right]$
- Estimation: compare each dose group to untreated ($d = 0$) separately, trace out the dose-response curve, then regress $\widehat{ATT}(d \mid d)$ on d
- Requires "strong" parallel trends: untreated potential outcomes trend the same across *all* dose groups
- R: `did` (extended); Avoid pooling all dose levels into a single TWFE regression!

Summary: Choosing Your DiD Estimator

- **Standard DiD / TWFE**: 2 groups, 2 periods, homogeneous effects
- **Callaway & Sant'Anna / Sun & Abraham**: staggered binary treatment, heterogeneous effects
- **Factorial DiD**: when events affect all units; distinguishes effect modification from causal moderation
- **Callaway, Goodman-Bacon, & Sant'Anna**: continuous/multi-valued treatment
- **Reanalysis evidence**: modern methods usually confirm, but sometimes overturn, published findings
- Always think about: What comparisons is your estimator making? Are those comparisons valid?